

Valoquent: Real-Time Video Conversation with History as an Adaptive Learning Instrument

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Abstract

During successful communication a listener’s neural activity comes to mirror the speaker’s, and the tighter that coupling, the better the listener understands [Stephens et al. 2010]; face-to-face dialogue produces a component of this synchrony that the same conversation without the visible face does not [Jiang et al. 2012]. In childhood, back-and-forth conversational turns predict language-related brain development more than passive exposure does [Romeo et al. 2018]. Conversation is, in a measurable sense, a mechanism for learning, and Valoquent runs that mechanism for two kinds of learning at once: a user holds a real-time video conversation with a figure from history and learns that figure’s life and ideas while learning the craft of quality conversation itself. A foundation language model generates each figure’s text; the contribution is the orchestration around it. The Conversation Meter reads twenty-two behavioral signals on every turn and shows the speaker a running measure of conversation quality, the score drives a two-layer disclosure system that rewrites the figure’s prompt while the session is still running, and a cross-session memory architecture carries what each conversation earned into the next. The format does the rest of the work: the personas are grounded in the historical record and rendered as stylized, lip-synced avatars, so what a user learns is history rather than invention, and the video is real time, so the exchange is genuine practice that cannot be paused, looked up, or rehearsed. In calibration, language-model-driven user archetypes hold hundreds of simulated sessions with the figures, and the meter separates engaged learners from shallow enthusiasm and sustained provocation.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Natural language processing*; • **Social and professional topics** → *Computing education*.

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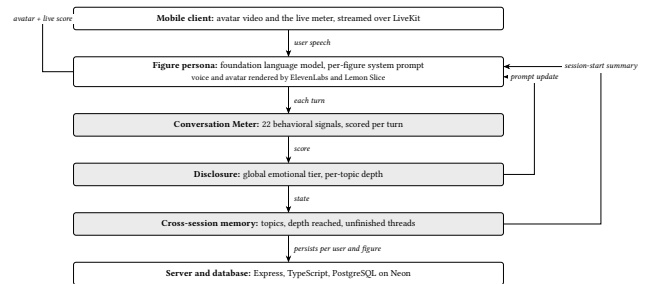


Figure 1: Valoquent’s data flow. Each turn, the figure’s reply and the user’s response reach the Conversation Meter; the score drives the disclosure level, which rewrites part of the figure’s prompt mid-session, and state persists in cross-session memory. Shaded components are the in-house contribution. The score is shown to the speaker, not to an observer.

1 Introduction

The motivation is access, and the gap runs through the lifespan, not only childhood. The amount of language a child hears varies widely by family [Hart and Risley 1995], back-and-forth conversational turns predict language development [Romeo et al. 2018], and children’s substantive questions to adults fall sharply by age ten [Engel 2015]. In a 2025 survey of 2,117 US adults, the most common answer to how often they discuss a given public issue was “never” [Kesintürk et al. 2025]. Substantive conversation, empirically distinct from small talk and tied to well-being [Mehl et al. 2010], stays gated by class, geography, and family circumstance [Lareau 2003]. Valoquent provides a substantive interlocutor, drawn from history, on demand.

2 System Architecture

A session looks like this: a user asks Marie Curie why she and Pierre refused to patent the radium process; the meter reads the question’s grounding and depth and adjusts the score; the disclosure layer opens her scientific-work topic one level; and her next answer draws on material from her own writings that a flatter question would not have reached. Underneath that loop, Valoquent runs as a React Native and Expo mobile client, an Express and TypeScript server with PostgreSQL persistence on Neon, and a Python conversation agent that carries the scoring, memory, and disclosure logic. Figure 1 shows the data flow.

2.1 Avatar and Voice Pipeline

Each figure appears as a real-time, lip-synced avatar that Lemon Slice generates from a single portrait with no per-figure training, streamed to the client over LiveKit, and stylized rather than photorealistic. Mori recommended a deliberately nonhuman design over more faithful rendering, since a figure that approaches human likeness without reaching it reads as eerie rather than appealing [Mori 2012]; more recent work shows the choice also helps learning. Across pedagogical agents at four levels of realism, cartoon agents produced the best quiz performance and realistic ones the worst [Raman et al. 2024], consistent with near-human realism taxing working memory the task itself needs [Sweller 1988], and work extending the uncanny valley to simulations found that mismatched realism cues worsen learning [Howard 2017]. ElevenLabs handles voice synthesis, speech recognition, and retrieval-augmented generation over each figure’s knowledge base. A foundation language model generates the figure’s text, conditioned by a per-character system prompt. The orchestration layer around that model is where Valoquent’s own work lives.

2.2 Character Personas

The figures are drawn from history and span philosophy, science, and the fight for justice, among them Marcus Aurelius, Marie Curie, and Frederick Douglass. Each persona is defined by its own system prompt and calibrated against that figure’s own writings, Curie against her radium memoir, Douglass against his Narrative, so each voice has a documented source rather than a generic one. The discipline matters because language models drift from assigned personas as a conversation runs [Abdulhai et al. 2025], and per-figure prompts grounded in source material are how distinct voices are preserved.

2.3 Memory

Each figure remembers the user across sessions. The design follows Clark’s account of conversation as the steady accumulation of common ground [Clark 1996]: the agent keeps explicit state about what has been shared and uses it to decide what to do next. Memory is stored per user and per figure: topics, depth reached, threads left unfinished. A compact summary is injected into the system prompt when a session begins, and per-topic disclosure state carries forward, so a user resumes at the depth they reached while subjects they never raised stay closed. This state has a user-facing surface: each figure’s Tapestry, a constellation map of the topics, events, and turning points of that figure’s life, where nodes start dark and light up as conversation reaches them. Within a live session, the agent keeps a turn-by-turn record of topic, depth, signals, and disclosure state, and uses it to rewrite parts of the figure’s prompt as the conversation moves.

3 The Conversation Meter

The Conversation Meter scores conversation quality turn by turn. It reads each pair of figure utterance and user response, produces a signed adjustment, and maintains a running score the user sees as one of five named zones, from Surface Skim to Flow State, after Csikszentmihalyi’s account of flow [Csikszentmihalyi 1990].

Dialogue evaluation has identified which automated signals predict human quality judgment [Deriu et al. 2021]; those scores live with researchers and analytics dashboards. The meter is built instead as a real-time, speaker-facing instrument. Its premise, that conversation quality is measurable while the conversation is still happening, follows from Stephens et al. [2010]: successful exchange registers physically, as speaker-listener neural coupling, while it unfolds. The meter does not grade the user. It shows them where the conversation is while they are still in it.

3.1 What the Meter Reads

The meter reads twenty-two behavioral signals from each turn, grouped into five families: **topicality and grounding** (whether the user is engaged with the figure’s substance and the prior turn); **substance and depth** (intellectual weight, scored against Bloom’s taxonomy [Bloom 1956] from recall through analysis to synthesis and evaluation); **engagement markers** (humor, enthusiasm, sustained flow, and optional client-side vocal prosody); **adversarial pattern detection** (separating constructive challenge from sustained provocation); and **figure-side responsiveness**. Follow-up coherence is weighted heavily, since Huang et al. [2017] found follow-up questions to be the question type most strongly tied to a partner perceiving responsiveness. The selection and weighting of the signals draw on dialogue evaluation, conversation analysis, and learning science.

3.2 Hybrid Scoring

Heuristic signal detection runs on its own, and an optional language-model scorer can be blended in to judge each utterance pair as a whole, following Ghazarian et al. [2020], who showed that learned per-utterance engagement scores improve automatic metrics’ agreement with human quality judgment. The blend is stage-aware: early turns lean on heuristics, and the model’s share rises as context accumulates.

3.3 Two-Layer Disclosure

The score drives a two-layer disclosure system that decides how much of itself a figure will reveal. The first layer is a global emotional tier: **Guarded** (factual, intellectual responses), **Warning** (professional struggles and considered opinions), **Open** (failures, doubts, and contested positions), and **Vulnerable** (full personal disclosure, without the cover of wit). The second layer is per-topic depth. Each figure defines several topic domains, and each moves independently through four levels, from surface to candid to deep to raw. The structure draws on work on intimacy: Aron et al. [1997] showed that closeness can be built through a graduated sequence of increasingly personal exchanges, and Altman and Taylor’s social penetration theory [Altman and Taylor 1973] describes intimacy as growing along both depth and breadth. Valoquent borrows the graduated structure but not the mutual self-disclosure: the user contributes intellectual engagement on the topic at hand, and what advances through the tiers is the figure’s documented historical material, not a profile of the user. The meter that gates the unlock is visible to the speaker throughout, and advancement depends on behavioral evidence.

3.4 Adversarial Engagement

Valoquent treats debate as a form of engagement rather than disruption. A user who argues that a figure is wrong about determinism is participating in the conversation, and the meter is built to register that: it separates constructive challenge, which is on topic and responsive, from trolling, which is challenge with no follow-through, using a behavioral pattern detector. The choice has empirical support: in a randomized classroom trial, a human-supervised LLM tutor built on Socratic questioning rather than answer-giving improved students' later performance on novel problems [LearnLM Team and Eedi 2025].

4 Calibration

I calibrate the meter against a library of language-model-driven user archetypes spanning engaged learners, shallow enthusiasts, and adversarial behavior patterns. Each archetype talks with each figure and carries an expected outcome: a curious student should finish engaged, a troll should not. Across hundreds of calibration runs, the share of core scenarios ending in their expected outcome rose from roughly 50 percent to 98 percent. A broader sweep still has a gap, concentrated in adversarial archetypes whose debate language slips past keyword-based topicality checks, and a challenge-detection module is planned to close it. Calibration is aimed at conversational realism, not the pass rate itself. Real conversationalists calibrate the shape of each turn to the moment [Sacks et al. 1974], so a meter tuned to a single ideal shape would measure the wrong thing.

5 Post-Session Learning

Two post-session features carry a conversation past the moment it happened. A Learning Report is a personalized PDF, generated through Foxit, gathering the session summary, key facts, and questions for the next conversation. After an engaged session, the user is also prompted to record a short spoken reflection on what they learned; it is transcribed and scored for which topics they retained. That step reinforces learning through retrieval practice [Roediger and Karpicke 2006]. It also hedges against the cognitive-crutch effect, where AI assistance during learning can reduce later retention [Barcaui 2025].

6 Discussion and Future Work

The premise behind Valoquent is that one activity can teach two things at once: the substance a historical figure carries, and the craft of conversation itself. The real-time video format is what holds both together. The meter, the disclosure tiers, and the memory architecture are how that dual learning is delivered and made visible. Valoquent is live on the App Store and Google Play.

Real limits remain. Adversarial archetypes can still score too high when their language passes topicality checks, vocal prosody is not yet enabled in production, and simulation-based calibration needs human evaluation alongside it. The work ahead includes classroom deployment with a teacher-facing view, on-device vocal analysis, and reports that synthesize a learner's exploration across several figures.

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